**OSMANIA UNIVERSITY**
**FACULTY OF ENGINEERING**
**UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)**
**B.E. ( CSE,AI&ML ) III-Semester (MAIN) Examinations DEC-2025/ JAN-2026**

# DATA STRUCTURES

**Time : 3 hours** **Max. Marks : 60**

**Note :** *i) Answer **Question No. 1** (Compulsory) and answer any **four questions** from the remaining questions (2- 7). Each question caries 12 marks.*
*ii) Answers must be written in same order as they occur in the Question Paper.*
*iii) **Missing data, if any may suitably be assumed.***

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|---|---|---|---|---|---|
| 1. | a) | Define an algorithm and list the characteristics of an algorithm. | 2 | 1 | 1 |
| | b) | State the applications of stack. | 2 | 2 | 2 |
| | c) | Find the maximum and minimum number of children a node can have in a B-Tree of order 7. | 2 | 3 | 2 |
| | d) | Give an example to traverse a graph using DFS. | 2 | 2 | 2 |
| | e) | What is the Worst Case time complexity of quick sort? Specify the nature of input to get Worst Case complexity. | 2 | 3 | 3 |
| | f) | How is Asymptotic Notation useful? | 2 | 2 | 1 |
| 2. | a) | Differentiate between Singly Linked Lists and Doubly Linked Lists with suitable examples. Which is better in terms of time complexity? | 7 | 2 | 1 |
| | b) | How is sparse matrix represented? | 5 | 2 | 2 |
| 3. | a) | Explain the various collision resolving techniques used in hash table representation with suitable examples. | 6 | 3 | 2 |
| | b) | Write a procedure to implement queue using linked list. | 6 | 3 | 2 |
| 4. | a) | Construct an AVL tree by specifying type of the rotation for the following data: 40 50 60 15 18 42 25 98 62 46 65 | 6 | 3 | 2 |
| | b) | Explain the Operations on B-Trees with suitable examples. | 6 | 2 | 2 |
| 5. | a) | Write an algorithm to find the degree, in degree and out degree of a node and number of edges a in a graph when graph is represented as matrix. | 6 | 3 | 2 |
| | b) | Explain the procedure to find the minimum cost spanning tree of an undirected weighted graph with an example. | 6 | 2 | 2 |


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|---|---|---|---|---|---|
| 6. | a) | Write an algorithm to sort the elements using Heap sort algorithm, draw the heap tree and find the content of array after each iteration if the initial array content is 54 26 93 17 77 31 44 55 20. | 7 | 3 | 3 |
| | b) | What are the applications of heap trees? | 5 | 2 | 3 |
| 7. | a) | Show the application of stack for postfix evaluation with suitable example. | 6 | 2 | 2 |
| | b) | Illustrate the operations of binary search tree with examples. | 6 | 2 | 2 |

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